A. Notation table

Table 1. Nomenclature

Symbol	Description
1. Ground MDP Components	
$\mathcal{M}_{\mathcal{S}}$	The ground Markov Decision Process (MDP).
$\mathcal{S}$	The ground state space.
$\mathcal{A}$	The action space.
$P_{\mathcal{SA}}$ (or $P^a_{\mathcal{S}}$)	Ground transition probability matrix, $P(s'\|s,a)$.
$r(s,a)$	Reward function for transitioning from (s,a).
γ	The discount factor, $\gamma \in (0,1)$.
$R_{\max}$	Maximum absolute reward, $\|r(s,a)\| \leq R_{\max}$.
$\xi_{\mathcal{S}}$	Initial state distribution for the ground MDP.
2. Abstraction Components	
U	The finite abstract state space.
$\nu(s\|u)$	Encoding distribution: probability of ground state s given abstract state u.
P_ν	The $\|U\| \times \|\mathcal{S}\|$ encoding (row-stochastic) matrix.
$P_\nu^\dagger$	Moore-Penrose pseudoinverse of the encoding matrix P_ν.
$r_U(u,a)$	Abstract reward function, $\sum_s \nu(s\|u) r(s,a)$.
$P^a_{U,\nu}$	Abstract transition matrix for action a, $P_\nu P^a_{\mathcal{S}} P_\nu^\dagger$.
$\mathcal{G}^\nu$	Bipartite support graph between U and $\mathcal{S}$.
$\alpha_{i,j}$	Elementary transition vectors $P_{\mathcal{SA}}(\cdot\|s_i,a_j)$.
$\mathcal{F}$	Maximal linearly independent subset (basis) of transition vectors.
3. Policies and Value Functions	
π	A policy $\pi(a\|s)$.
$\Pi_{\mathcal{S}}$	The set of all policies on the ground state space $\mathcal{S}$.
Π_U	The set of all policies on the abstract state space U.
$\pi^\star$	The optimal policy in the ground MDP $\mathcal{M}_{\mathcal{S}}$.
$f_\nu(\pi)$	The abstract policy induced by π via f_ν.
$\pi^\star_\nu$	The abstract optimal policy, $f_\nu(\pi^\star)$.
$\widehat{\pi}$	The empirical (approximate) greedy policy computed on U.
$\widehat{\pi}^\uparrow$	The lifted policy, where $\widehat{\pi}^\uparrow(a\|s) := \widehat{\pi}(a\|u)$.
$\Pi_{\mathcal{S}}^{\mathcal{G}^\nu}$	Set of policies constant on components of $\mathcal{G}^\nu$.
$V^\pi_{\mathcal{S}}$	State value function for policy π in the ground MDP.
$Q^\pi_{\mathcal{S}}$	State-action value function for π in the ground MDP.
$V_U^{f_\nu(\pi)}$	State value function for the abstract policy $f_\nu(\pi)$.
$J_{\mathcal{S}}(\pi)$	Performance (expected discounted return) of π in $\mathcal{M}_{\mathcal{S}}$.
$\mathcal{T}_U, \widehat{\mathcal{T}}_U$	The true and empirical Bellman operators on U.
4. Sample-Based Estimation and Errors	
N	Number of sample trajectories.
H	Length of sample trajectories.
$n(u,a)$	Sample counts for abstract state-action pair (u,a).
$n_{\min}$	Minimum sample count, $\min_{u,a} n(u,a)$.
$\widehat{r}_U$	Empirical estimate of the abstract reward r_U.
$\widehat{P}^a_{U,\nu}$	Empirical estimate of the abstract transition $P^a_{U,\nu}$.
ε_{rew}	Reward estimation error $\max_{u,a} \|\widehat{r}_U - r_U\|$.
ε_{mdl}	Model (transition) estimation error $\max_{u,a} \mathrm{TV}(\widehat{P}^a, P^a)$.
ε_{abs}	Abstraction error (homomorphism violation).
ε_{com}	Commutativity (policy) error.
ε_{opt}	Numerical tolerance error from finite inner solves.
ϵ	The target ϵ-optimality gap for performance.
δ	The probability of failure (i.e., not achieving ϵ-optimality).

B. A Data-Dependent Bound for Epoch-Based H-UCB under Hard Encodings

In this appendix, we analyze a special case of adaptive H-UCB in which the encoding is estimated from data at the beginning of each epoch and then kept fixed within the epoch. The goal is to make explicit how the structural abstraction term depends on the number of samples used to estimate the encoding.

Our purpose here is not to claim a fully general regret guarantee for arbitrary online abstraction updates. Instead, we show that under a natural hard-aggregation setting, one can derive a concrete data-dependent bound of the form

$$\varepsilon_{\mathrm{abs}}^{(k)} \lesssim \mathcal{L}_k^\star + \sqrt{\frac{\log(1/\delta)}{m_k}},$$

which can then be plugged directly into the epoch-wise regret decomposition used in the main text.

B.1. Setup and additional assumptions

We consider deterministic hard encodings of the ground state space $\mathcal{S}$ into a finite abstract space $\mathcal{U}$. Let

$$\phi : \mathcal{S} \to \mathcal{U}$$

be a deterministic encoding, and let

$$C_u(\phi) := \{s \in \mathcal{S} : \phi(s) = u\}$$

denote the corresponding cell for $u \in \mathcal{U}$.

For each such ϕ, we define the hard encoding matrix P_{ν_ϕ} by

$$P_{\nu_\phi}(s \mid u) = \frac{1}{|C_u(\phi)|}\mathbf{1}\{s \in C_u(\phi)\}. \tag{1}$$

Thus, given an abstract state u, the decoder $P_{\nu_\phi}(\cdot \mid u)$ is uniform over the ground states in the cell $C_u(\phi)$.

We also define the canonical abstraction map

$$P_{\nu_\phi}^\dagger(u \mid s) := \mathbf{1}\{\phi(s) = u\},$$

so that $P_{\nu_\phi}^\dagger P_{\nu_\phi}$ is the block-averaging projection induced by ϕ.

Assumption B.1 (Cell-consistent policy class)**.** For a fixed deterministic encoding ϕ, define

$$\Pi_{\mathcal{S}}^{G_\phi} := \Big\{\pi : \ \pi(\cdot \mid s) = \pi(\cdot \mid s') \text{ whenever } \phi(s) = \phi(s')\Big\}.$$

For every $\pi \in \Pi_{\mathcal{S}}^{G_\phi}$, define the abstract policy $f_\phi(\pi)$ by

$$f_\phi(\pi)(a \mid u) := \pi(a \mid s), \qquad s \in C_u(\phi). \tag{2}$$

This is well-defined by construction.

Assumption B.2 (Epoch-wise frozen encoding)**.** Time is partitioned into epochs

$$1 = \tau_1 < \tau_2 < \cdots < \tau_{K+1} = T + 1, \qquad T_k := \tau_{k+1} - \tau_k.$$

At the beginning of epoch k, the algorithm selects an encoding $\widehat{\phi}_k$, constructs the associated hard encoding $P_{\nu_{\widehat{\phi}_k}}$, and keeps it fixed throughout epoch k.

Assumption B.3 (Ground-state coverage data)**.** For each epoch k, before epoch k begins, the learner has access to a dataset $\mathcal{D}_k$ collected from the ground MDP. For each action $a \in \mathcal{A}$, let $\widehat{P}_{\mathcal{S},k}^a$ denote the empirical ground transition kernel estimated from $\mathcal{D}_k$.

Assume that each ground state-action pair (s, a) has been observed at least m_k times in $\mathcal{D}_k$.

Remark. Assumption B.3 is strong, but it isolates the statistical question of *estimating the encoding from data*. The resulting bound should be interpreted as a clean special case rather than a complete analysis of fully adaptive H-UCB.

B.2. A data-dependent inconsistency loss for hard encodings

For a deterministic encoding ϕ, define the induced abstract transition kernel

$$P^a_{\mathcal{U},\phi} := P_{\nu_\phi} P^a_{\mathcal{S}} P^\dagger_{\nu_\phi}. \tag{3}$$

Similarly, using the empirical ground kernel $\widehat{P}^a_{\mathcal{S},k}$, define

$$\widehat{P}^a_{\mathcal{U},\phi,k} := P_{\nu_\phi} \widehat{P}^a_{\mathcal{S},k} P^\dagger_{\nu_\phi}. \tag{4}$$

We measure the quality of an encoding ϕ via the one-step abstraction inconsistency

$$\mathcal{L}(\phi) := \max_{u\in\mathcal{U},\, a\in\mathcal{A}} \left\| \left(P^a_{\mathcal{U},\phi} P_{\nu_\phi} - P_{\nu_\phi} P^a_{\mathcal{S}} \right)_{u,\cdot} \right\|_1, \tag{5}$$

and its empirical analogue

$$\widehat{\mathcal{L}}_k(\phi) := \max_{u\in\mathcal{U},\, a\in\mathcal{A}} \left\| \left(\widehat{P}^a_{\mathcal{U},\phi,k} P_{\nu_\phi} - P_{\nu_\phi} \widehat{P}^a_{\mathcal{S},k} \right)_{u,\cdot} \right\|_1. \tag{6}$$

Given a candidate class Φ_k of encodings at epoch k, we define the empirical encoding estimator by

$$\widehat{\phi}_k \in \arg\min_{\phi\in\Phi_k} \widehat{\mathcal{L}}_k(\phi). \tag{7}$$

B.3. Uniform estimation of the encoding loss

The first result shows that the empirical inconsistency loss uniformly approximates its population counterpart.

Proposition B.4 (Uniform estimation of the hard-encoding inconsistency loss)**.** *Fix an epoch k, and define*

$$\eta_k := \max_{s\in\mathcal{S},\, a\in\mathcal{A}} \left\| \left(\widehat{P}^a_{\mathcal{S},k} - P^a_{\mathcal{S}} \right)_{s,\cdot} \right\|_1. \tag{8}$$

Then for every deterministic hard encoding ϕ,

$$\left| \widehat{\mathcal{L}}_k(\phi) - \mathcal{L}(\phi) \right| \le 2\eta_k. \tag{9}$$

Consequently, if $\widehat{\phi}_k$ is defined by (7), then

$$\mathcal{L}(\widehat{\phi}_k) \le \mathcal{L}^\star_k + 4\eta_k, \qquad \mathcal{L}^\star_k := \min_{\phi\in\Phi_k} \mathcal{L}(\phi). \tag{10}$$

Proof. Fix an encoding ϕ, an abstract state $u \in \mathcal{U}$, and an action $a \in \mathcal{A}$. Define

$$\Delta^a_\phi := P^a_{\mathcal{U},\phi} P_{\nu_\phi} - P_{\nu_\phi} P^a_{\mathcal{S}}, \qquad \widehat{\Delta}^a_{\phi,k} := \widehat{P}^a_{\mathcal{U},\phi,k} P_{\nu_\phi} - P_{\nu_\phi} \widehat{P}^a_{\mathcal{S},k}.$$

Then

$$\mathcal{L}(\phi) = \max_{u,a} \|(\Delta^a_\phi)_{u,\cdot}\|_1, \qquad \widehat{\mathcal{L}}_k(\phi) = \max_{u,a} \|(\widehat{\Delta}^a_{\phi,k})_{u,\cdot}\|_1.$$

Using the definitions (3) and (4), we have

$$\widehat{P}^a_{\mathcal{U},\phi,k} - P^a_{\mathcal{U},\phi} = P_{\nu_\phi} \big(\widehat{P}^a_{\mathcal{S},k} - P^a_{\mathcal{S}} \big) P^\dagger_{\nu_\phi}.$$

Therefore,

$$\begin{aligned} \widehat{\Delta}^a_{\phi,k} - \Delta^a_\phi &= \big(\widehat{P}^a_{\mathcal{U},\phi,k} - P^a_{\mathcal{U},\phi} \big) P_{\nu_\phi} - P_{\nu_\phi} \big(\widehat{P}^a_{\mathcal{S},k} - P^a_{\mathcal{S}} \big) \\ &= P_{\nu_\phi} \big(\widehat{P}^a_{\mathcal{S},k} - P^a_{\mathcal{S}} \big) \big(P^\dagger_{\nu_\phi} P_{\nu_\phi} - I \big). \end{aligned}$$

Let

$$M_k^a := \widehat{P}_{\mathcal{S},k}^a - P_{\mathcal{S}}^a.$$

Since each row of P_{ν_ϕ} is a probability distribution over the corresponding cell, taking a convex combination cannot increase the row-wise ℓ_1 norm. Hence, for every u,

$$\|(P_{\nu_\phi} M_k^a)_{u,\cdot}\|_1 \le \max_{s\in\mathcal{S}} \|(M_k^a)_{s,\cdot}\|_1 \le \eta_k.$$

Likewise, under the hard-aggregation construction, $P_{\nu_\phi}^\dagger P_{\nu_\phi}$ is the block-averaging projection associated with ϕ, and thus does not increase row-wise ℓ_1 norms. Therefore,

$$\|(P_{\nu_\phi} M_k^a P_{\nu_\phi}^\dagger P_{\nu_\phi})_{u,\cdot}\|_1 \le \eta_k.$$

Applying the triangle inequality yields

$$\|(\widehat{\Delta}_{\phi,k}^a - \Delta_\phi^a)_{u,\cdot}\|_1 \le 2\eta_k.$$

Taking the maximum over u and a proves (9).

Now let

$$\phi_k^\star \in \arg\min_{\phi\in\Phi_k} \mathcal{L}(\phi).$$

Since $\widehat{\phi}_k$ minimizes the empirical loss,

$$\widehat{\mathcal{L}}_k(\widehat{\phi}_k) \le \widehat{\mathcal{L}}_k(\phi_k^\star).$$

Using (9) twice, we obtain

$$\mathcal{L}(\widehat{\phi}_k) \le \widehat{\mathcal{L}}_k(\widehat{\phi}_k) + 2\eta_k \le \widehat{\mathcal{L}}_k(\phi_k^\star) + 2\eta_k \le \mathcal{L}(\phi_k^\star) + 4\eta_k = \mathcal{L}_k^\star + 4\eta_k,$$

which proves (10). □

The next lemma converts the empirical ground-model error η_k into an explicit sample-size bound.

Lemma B.5 (Ground-model concentration under coverage)**.** *Under Assumption B.3, there exists a universal constant $c > 0$ such that for every epoch k and every $\delta_k \in (0,1)$,*

$$\Pr\left(\eta_k \le c\sqrt{\frac{|\mathcal{S}| + \log\left(\frac{|\mathcal{S}||\mathcal{A}|}{\delta_k}\right)}{m_k}}\right) \ge 1 - \delta_k.$$

Consequently, with probability at least $1-\delta_k$,

$$\mathcal{L}(\widehat{\phi}_k) \le \mathcal{L}_k^\star + 4c\sqrt{\frac{|\mathcal{S}| + \log\left(\frac{|\mathcal{S}||\mathcal{A}|}{\delta_k}\right)}{m_k}}. \tag{11}$$

Proof. For each fixed (s,a), the row

$$\widehat{P}_{\mathcal{S},k}^a(\cdot \mid s)$$

is the empirical frequency vector of a multinomial distribution over $|\mathcal{S}|$ outcomes based on at least m_k samples. Standard ℓ_1 concentration for multinomial empirical distributions implies that, for some universal constant $c > 0$,

$$\Pr\left(\left\|\left(\widehat{P}_{\mathcal{S},k}^a - P_{\mathcal{S}}^a\right)_{s,\cdot}\right\|_1 \le c\sqrt{\frac{|\mathcal{S}| + \log\left(\frac{|\mathcal{S}||\mathcal{A}|}{\delta_k}\right)}{m_k}}\right)$$

holds uniformly over all (s,a) after a union bound. This proves the stated bound on η_k.

Substituting this into Proposition B.4 immediately yields (11). □

B.4. Bridge to the abstraction error terms

We now relate the inconsistency loss $\mathcal{L}(\phi)$ to the abstraction quantities used in the main text.

Lemma B.6 (Bridge from one-step inconsistency to $(\varepsilon_{\mathrm{abs}}, \varepsilon_{\mathrm{com}})$)**.** *Fix a deterministic hard encoding ϕ, let P_{ν_ϕ} be defined by (1), and restrict the ground policy class to $\Pi_{\mathcal{S}}^{G_\phi}$ from Assumption B.1. Then:*

$$\varepsilon_{\mathrm{com}}(\phi) = 0, \tag{12}$$

and

$$\varepsilon_{\mathrm{abs}}(\phi) \le \mathcal{L}(\phi). \tag{13}$$

Proof. We first prove (12). By definition of $\Pi_{\mathcal{S}}^{G_\phi}$, whenever $\phi(s) = \phi(s')$, we have

$$\pi(\cdot \mid s) = \pi(\cdot \mid s').$$

Hence the lifted abstract policy $f_\phi(\pi)$ is well-defined by (2), and for any $u \in \mathcal{U}$, any $s \in C_u(\phi)$, and any $a \in \mathcal{A}$,

$$f_\phi(\pi)(a \mid u) = \pi(a \mid s).$$

Therefore the action distributions match exactly on each cell, so the commutativity mismatch vanishes:

$$\sum_{a \in \mathcal{A}} |f_\phi(\pi)(a \mid u) - \pi(a \mid s)| = 0.$$

Taking the supremum over all $\pi \in \Pi_{\mathcal{S}}^{G_\phi}$, $u \in \mathcal{U}$, and $s \in C_u(\phi)$ yields (12).

We next prove (13). Fix any $\pi \in \Pi_{\mathcal{S}}^{G_\phi}$ and let $\mu := f_\phi(\pi)$. By cell consistency, the action probabilities in each cell are constant, so

$$P_{\mathcal{U}}^{\mu} = \sum_{a \in \mathcal{A}} \Lambda_a^{\mu} P_{\mathcal{U},\phi}^{a}, \qquad P_{\mathcal{S}}^{\pi} = \sum_{a \in \mathcal{A}} \Lambda_a^{\pi} P_{\mathcal{S}}^{a},$$

where the diagonal action-selection operators agree cell-wise. Therefore,

$$P_{\mathcal{U}}^{\mu} P_{\nu_\phi} - P_{\nu_\phi} P_{\mathcal{S}}^{\pi} = \sum_{a \in \mathcal{A}} \Lambda_a^{\mu} \big(P_{\mathcal{U},\phi}^{a} P_{\nu_\phi} - P_{\nu_\phi} P_{\mathcal{S}}^{a} \big).$$

For any abstract state u, taking the u-th row and applying the triangle inequality gives

$$\begin{aligned} \left\| \big(P_{\mathcal{U}}^{\mu} P_{\nu_\phi} - P_{\nu_\phi} P_{\mathcal{S}}^{\pi} \big)_{u,\cdot} \right\|_1 &\le \sum_{a \in \mathcal{A}} \mu(a \mid u) \left\| \big(P_{\mathcal{U},\phi}^{a} P_{\nu_\phi} - P_{\nu_\phi} P_{\mathcal{S}}^{a} \big)_{u,\cdot} \right\|_1 \\ &\le \max_{a \in \mathcal{A}} \left\| \big(P_{\mathcal{U},\phi}^{a} P_{\nu_\phi} - P_{\nu_\phi} P_{\mathcal{S}}^{a} \big)_{u,\cdot} \right\|_1 \\ &\le \mathcal{L}(\phi). \end{aligned}$$

Taking the supremum over all $\pi \in \Pi_{\mathcal{S}}^{G_\phi}$ and all $u \in \mathcal{U}$ yields (13). □

B.5. A data-dependent regret guarantee for epoch-based H-UCB

We now combine the preceding results with the fixed-abstraction regret analysis from the main text.

For convenience, define the per-epoch structural quantities

$$\varepsilon_{\mathrm{abs}}^{(k)} := \varepsilon_{\mathrm{abs}}(\widehat{\phi}_k), \qquad \varepsilon_{\mathrm{com}}^{(k)} := \varepsilon_{\mathrm{com}}(\widehat{\phi}_k).$$

Theorem B.7 (Epoch-based H-UCB with data-estimated hard encodings)**.** *Consider the epoch-based adaptive H-UCB procedure under Assumptions B.1–B.3. In epoch k, the encoding $\widehat{\phi}_k$ is selected according to (7), the induced hard encoding $P_{\nu_{\widehat{\phi}_k}}$ is kept fixed throughout the epoch, and the inner loop runs the fixed-abstraction Abstract-UCB procedure using only epoch-local statistics.*

Let $\mathcal{F}_{\tau_k-1}$ denote the sigma-field generated by all randomness available before epoch k begins. Assume further that for every epoch k, every confidence level $\beta_k \in (0,1)$, and every $\mathcal{F}_{\tau_k-1}$-measurable encoding ϕ that is frozen throughout epoch k, the conclusion of Theorem 4.4 applies conditionally on $\mathcal{F}_{\tau_k-1}$ when the inner loop is run on epoch k with epoch-local UCB statistics and confidence parameter β_k. Then, for any confidence allocations $(\alpha_k)_{k=1}^K$ and $(\beta_k)_{k=1}^K$ satisfying

$$\sum_{k=1}^{K}(\alpha_k + \beta_k) \le \delta,$$

with probability at least $1-\delta$,

$$\begin{aligned}\mathrm{Regret}(T) \le\; & C\frac{R_{\max}}{(1-\gamma)^3}\sum_{k=1}^{K}\sqrt{|\mathcal{U}||\mathcal{A}|\,T_k\log\Big(\frac{|\mathcal{U}||\mathcal{A}|T}{\beta_k}\Big)} \\ & + \frac{2\gamma R_{\max}}{(1-\gamma)^2}\sum_{k=1}^{K}T_k\left[\mathcal{L}_k^\star + 4c\sqrt{\frac{|\mathcal{S}| + \log\big(\frac{|\mathcal{S}||\mathcal{A}|}{\alpha_k}\big)}{m_k}}\right],\end{aligned} \tag{14}$$

where

$$\mathcal{L}_k^\star := \min_{\phi\in\Phi_k}\mathcal{L}(\phi),$$

and $c>0$ is the universal constant from Lemma B.5.

In particular, if

$$\alpha_k = \beta_k = \frac{\delta}{2K}, \qquad k = 1,\dots,K,$$

then with probability at least $1-\delta$,

$$\begin{aligned}\mathrm{Regret}(T) \le\; & C\frac{R_{\max}}{(1-\gamma)^3}\sum_{k=1}^{K}\sqrt{|\mathcal{U}||\mathcal{A}|\,T_k\log\Big(\frac{2K|\mathcal{U}||\mathcal{A}|T}{\delta}\Big)} \\ & + \frac{2\gamma R_{\max}}{(1-\gamma)^2}\sum_{k=1}^{K}T_k\left[\mathcal{L}_k^\star + 4c\sqrt{\frac{|\mathcal{S}| + \log\big(\frac{2K|\mathcal{S}||\mathcal{A}|}{\delta}\big)}{m_k}}\right].\end{aligned} \tag{15}$$

Proof. Fix an epoch k, and define the pre-epoch filtration

$$\mathcal{F}_{\tau_k-1} := \sigma\big(\text{all randomness observed before epoch } k\big).$$

By construction, the dataset $\mathcal{D}_k$, the empirical kernel $\widehat{P}^a_{\mathcal{S},k}$, and hence the estimator $\widehat{\phi}_k$ are all $\mathcal{F}_{\tau_k-1}$-measurable.

Let

$$E_k^{\mathrm{enc}} := \left\{\mathcal{L}(\widehat{\phi}_k) \le \mathcal{L}_k^\star + 4c\sqrt{\frac{|\mathcal{S}| + \log\big(\frac{|\mathcal{S}||\mathcal{A}|}{\alpha_k}\big)}{m_k}}\right\}.$$

By Lemma B.5,

$$\Pr\big(E_k^{\mathrm{enc}}\big) \ge 1-\alpha_k. \tag{16}$$

On the event E_k^{enc}, Lemma B.6 yields

$$\varepsilon_{\mathrm{com}}^{(k)} = 0, \qquad \varepsilon_{\mathrm{abs}}^{(k)} \le \mathcal{L}_k^\star + 4c\sqrt{\frac{|\mathcal{S}| + \log\big(\frac{|\mathcal{S}||\mathcal{A}|}{\alpha_k}\big)}{m_k}}.$$

Now condition on $\mathcal{F}_{\tau_k-1}$. Since $\widehat{\phi}_k$ and $P_{\nu_{\widehat{\phi}_k}}$ are $\mathcal{F}_{\tau_k-1}$-measurable and remain fixed throughout epoch k, the epoch-k inner loop is exactly a frozen-encoding instance of Abstract-UCB. Therefore, by the conditional version of Theorem 4.4 assumed in the statement, there exists an event $E_k^{\mathrm{ucb}} \in \mathcal{F}_{\tau_{k+1}-1}$ such that

$$\Pr\big(E_k^{\mathrm{ucb}} \mid \mathcal{F}_{\tau_k-1}\big) \ge 1-\beta_k \qquad \text{a.s.}, \tag{17}$$

and on the event E_k^{ucb},

$$\mathrm{Regret}_k \le C\frac{R_{\max}}{(1-\gamma)^3}\sqrt{|\mathcal{U}||\mathcal{A}|\,T_k\log\Big(\frac{|\mathcal{U}||\mathcal{A}|T}{\beta_k}\Big)} + \frac{2\gamma R_{\max}}{(1-\gamma)^2}T_k\,\varepsilon_{\mathrm{abs}}^{(k)},$$

where the commutativity term vanishes because $\varepsilon_{\mathrm{com}}^{(k)} = 0$.

Hence, on the intersection

$$E_k := E_k^{\mathrm{enc}} \cap E_k^{\mathrm{ucb}},$$

we have

$$\begin{aligned}\mathrm{Regret}_k \le\ & C\frac{R_{\max}}{(1-\gamma)^3}\sqrt{|\mathcal{U}||\mathcal{A}|\,T_k\log\Big(\frac{|\mathcal{U}||\mathcal{A}|T}{\beta_k}\Big)} \\ & + \frac{2\gamma R_{\max}}{(1-\gamma)^2}T_k\left[\mathcal{L}_k^\star + 4c\sqrt{\frac{|\mathcal{S}| + \log\big(\frac{|\mathcal{S}||\mathcal{A}|}{\alpha_k}\big)}{m_k}}\right].\end{aligned} \tag{18}$$

We next bound the failure probability of E_k. Using (16), (17), and the tower property,

$$\begin{aligned}\Pr(E_k^c) &\le \Pr\big((E_k^{\mathrm{enc}})^c\big) + \Pr\big(E_k^{\mathrm{enc}} \cap (E_k^{\mathrm{ucb}})^c\big) \\ &= \Pr\big((E_k^{\mathrm{enc}})^c\big) + \mathbb{E}\big[\mathbf{1}_{E_k^{\mathrm{enc}}}\Pr\big((E_k^{\mathrm{ucb}})^c \mid \mathcal{F}_{\tau_k-1}\big)\big] \\ &\le \alpha_k + \beta_k.\end{aligned}$$

Therefore,

$$\Pr\Big(\bigcap_{k=1}^K E_k\Big) \ge 1 - \sum_{k=1}^K(\alpha_k+\beta_k) \ge 1-\delta.$$

On the event $\bigcap_{k=1}^K E_k$, summing (18) over $k = 1,\dots,K$ gives (14). The specialized bound (15) follows by setting $\alpha_k = \beta_k = \delta/(2K)$ for all k. □

Corollary B.8 (Doubling schedule)**.** *Under the conditions of Theorem B.7, suppose additionally that:*

(i) $K = O(\log T)$*;*

(ii) $m_k \asymp T_k \asymp 2^{k-1}$*;*

(iii) $\delta_k = \delta/K$ *for all* k*.*

Then, with probability at least $1-\delta$*,*

$$\begin{aligned}\mathrm{Regret}(T) \le\ & \widetilde{O}\Big(\frac{R_{\max}}{(1-\gamma)^3}\sqrt{|\mathcal{U}||\mathcal{A}|\,KT}\Big) \\ & + \frac{2\gamma R_{\max}}{(1-\gamma)^2}\left(\sum_{k=1}^K T_k\mathcal{L}_k^\star + \widetilde{O}(\sqrt{KT\,|\mathcal{S}|})\right).\end{aligned} \tag{19}$$

Proof. Under the doubling schedule,

$$\sum_{k=1}^K\sqrt{T_k} \le \sqrt{K\sum_{k=1}^K T_k} = \sqrt{KT}.$$

Therefore the statistical UCB term satisfies

$$\sum_{k=1}^K\sqrt{|\mathcal{U}||\mathcal{A}|\,T_k\log\Big(\frac{|\mathcal{U}||\mathcal{A}|T}{\delta}\Big)} = \widetilde{O}\big(\sqrt{|\mathcal{U}||\mathcal{A}|\,KT}\big).$$

Likewise, since $m_k \asymp T_k$,

$$\sum_{k=1}^{K} T_k \sqrt{\frac{|\mathcal{S}| + \log\left(\frac{|\mathcal{S}||\mathcal{A}|K}{\delta}\right)}{m_k}} = \widetilde{O}\left(\sqrt{KT\,|\mathcal{S}|}\right).$$

Substituting these estimates into (15) proves (19). □

Interpretation. Theorem B.7 shows that, in this hard-encoding special case, the abstraction term can be decomposed into:

1. an approximation term $\mathcal{L}_k^\star$, measuring the best one-step abstraction inconsistency achievable within the candidate class Φ_k; and
2. a statistical term of order $m_k^{-1/2}$, explicitly controlled by the number of ground samples used to estimate the encoding.

Thus, unlike a purely conditional regret bound stated in terms of abstract error constants, the present result makes the dependence on encoding estimation from data explicit.

Scope of the result. This appendix does not analyze the fully interleaved version of H-UCB in which the encoding changes arbitrarily online while all past planning statistics are reused. Rather, it establishes a concrete and analyzable intermediate result: once the encoding is updated only at epoch boundaries and kept fixed within each epoch, one can combine an encoding-estimation argument with the fixed-abstraction regret analysis from the main text.